

Model Question Paper

TED (15)-6042

Reg. No.....

(REVISION 2015)

Signature.....

DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY

Communication Systems

(Maximum Marks: 100)

[Time: 3 hours]

PART – A

(Maximum Marks: 10)

I. Answer all the questions in one or two sentences. Each question carries 2 Marks.

1. State the function of wave guide .
2. Name the component of DBS.
3. State numerical aperture.
4. List the components of RFID.
5. Draw the symbol of PIN diode. (5x2=10)

PART – B

(Maximum Marks: 30)

II. Answer any *five* of the following questions. Each question carries 6 marks.

1. Describe the working Gunn diode.
2. Describe the bunching process.
3. List any six features of CDMA.
4. Describe GPS navigation system.
5. Describe the cable losses.
6. Draw the GSM architecture.
7. Draw and discuss the characteristics of PIN diode . (5x6=30)

PART – C
(Maximum Marks: 60)

Answer *one* full question from each module. Each full question carries 15 marks.

MODULE– I

- III.a)** With a neat diagram describe the operation of magnetron tube. (7)
b) Describe the microwave communication system with a neat diagram. (8)

OR

- IV.a)** Draw and describe the functional block of a microwave receiver. (9)
b) Describe different type of horn antenna. (6)

MODULE – II

- a) Describe FDMA with a neat diagram. (10)
b) Discuss geostationary satellite. (5)

OR

- V.a)** Describe GIS with diagram. (9)
b) List the applications of satellite. (6)

MODULE – III

- VII. a)** Illustrate different modes of optical fibre. (9)
b) Describe Acceptance angle and NA. (6)

OR

- VIII. a)** Illustrate the Avalanche diode with neat diagram . (9)
b) List any six applications of fibre optics in communication. (6)

MODULE – IV

- IX. a)** Describe the different handoff strategies used in mobile communication. (9)
b) Describe the operations of cellular network with a diagram. (6)

OR

- X. a)** List the features of Wi Max. (8)
b) Compare GSM and CDMA. (7)